

OSMAN ABDUL-FATAWU IDDRISU

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EDUCATION

Master of Philosophy, Public Health: Field Epidemiology and Applied Biostatistics Jan 2025 – Sep 2026
Kwame Nkrumah University of Science and Technology Kumasi, Ghana

- Thesis: “*Mathematical Modelling of HIV Transmission Dynamics in Ghana: A Data-Driven SEIR Model Simulation for Evaluating Testing, Treatment, and Prevention Strategies*”.

Bachelor of Science, Actuarial Science Aug 2018 – Sep 2022
Kwame Nkrumah University of Science and Technology Kumasi, Ghana

- Thesis: “*Comparative Analysis Between Bayesian Regression and Classical Multiple Linear Regression*”.

RESEARCH INTERESTS

Mathematical Modelling · Spatial and Spatiotemporal Statistics · Bayesian Statistics · Statistical Machine Learning

TEACHING EXPERIENCE

Graduate Student Instructor Jan 2026 – Apr 2026
Kwame Nkrumah University of Science and Technology Kumasi, Ghana

- Instructed graduate-level courses: Introduction to R for Data Analysis, Advanced Epidemiology, and MySQL for Database Development.
- Delivered lectures and hands-on training in statistical programming, data analysis, and database design using R and SQL.

Teaching Assistant Nov 2022 – Dec 2023
Kwame Nkrumah University of Science and Technology Kumasi, Ghana

- Delivered lectures for undergraduate courses (STAT 153, STAT 253, STAT 485, MATH 164, PHARM 493) to over 235 students.
- Contributed to improved student outcomes (average pass rate 87%).

Summer School Tutor Jun 2018 – Sep 2022
Visionary Global College Kumasi, Ghana

- Taught and graded quizzes in Core Mathematics, Elective Mathematics, Physics, Integrated Science, and Chemistry.

RESEARCH EXPERIENCE

Lead Data Analyst (Contract) Apr 2025 – Present
World Veterinary Association Brussels, Belgium

- Selected for a multi-country research initiative (30+ countries) on One Health and Planetary Health education.
- Conducting statistical analyses and co-authoring manuscripts for peer-reviewed publication.

Research Assistant Jan 2025 – Apr 2025
Kwame Nkrumah University of Science and Technology Kumasi, Ghana

- Applied GIS, Stata, and spatial statistics in R to analyze regional health data.
- Supported targeted public health interventions and facilitated R training for 200+ students.

PUBLICATIONS

Owusu, S. E., Iddrisu, O. A., Anugwom, G., & Onyeaka, H. (2026). *Association between sleep duration and depressive symptoms among U.S. Adults: A Cross-Sectional Analysis of National Health and Nutrition Examination Survey 2021–2023*. *Cureus*, 18(8), e114486. <https://doi.org/10.7759/cureus.114486> (**Peer-reviewed**)

Iddrisu, O. A., Abukari, H. S., & Siddiq, A. I. (2026). *Bayesian Spatiotemporal Small-Area Estimation of HIV Testing Uptake in Ghana, 2008–2022: Integrating Machine-Learning-Derived Geospatial Covariates with District-Level BYM2-RW1 Modelling of the Ghana Demographic and Health Surveys*. **Preprint**. <https://doi.org/10.64898/2026.07.31.26359204>

Iddrisu, O. A., Isma-il, A., Abukari, H. S., & Siddiq, A. I. (2026). *Geospatial Patterns of Cariogenic Feeding Practices Among Children Aged 6–23 Months in Ghana: A Spatial and Multilevel Analysis of the 2022 Ghana Demographic and*

Health Survey. *medRxiv Preprint*. <https://doi.org/10.64898/2026.08.06.26359897>

Siddiq, A. I., Iddrisu, O. A., Abubakar, H. S., Sowah, S. N. T., & Botchway, S. Q. (2026). *Socioeconomic inequalities and spatial clustering in HIV testing uptake in Ghana: A concentration index and geospatial analysis of the 2022 GDHS*. *medRxiv Preprint*. <https://doi.org/10.64898/2026.08.11.26358504>

Abubakar, H. S., Siddiq, A. I., & Iddrisu, O. A. (2026). Spatial and machine learning analysis of breast and cervical cancer screening uptake in Ghana: Evidence from the 2022 Ghana Demographic and Health Survey. *medRxiv Preprint*. <https://doi.org/10.64898/2026.08.15.26360510>

Iddrisu, O. A., Owusu-Sekyere, F., Siddick, A. H., Asiamah-Asare, B. K. Y., & Nyadamu, S. D. (2026). *Spatio-temporal distribution of HIV cases in Ghana: A 5-year regional assessment using routine surveillance data*. *medRxiv Preprint*. <https://doi.org/10.64898/2026.08.20.26360906>

Bayesian Geospatial Mapping and Hotspot Prediction of Childhood Anemia in Ghana. Manuscript in preparation.

CONFERENCE PRESENTATIONS

3rd Biennial Postgraduate Conference, College of Health Sciences, KNUST, Ghana (June 2026).

Presented: “*Socioeconomic Inequalities and Spatial Clustering in HIV Testing Uptake in Ghana: A Concentration Index and Geospatial Analysis of the 2022 GDHS*”.

IndabaX Data Science Summit, Ho Technical University, Ghana (June 2026).

Presented: “*Spatio-temporal Distribution of HIV in Ghana: A 5-Year Regional Assessment Using Routine Surveillance*”.

Abstract: <https://openreview.net/forum?id=s5r3RtW8hZ>

International Conference on AI in Healthcare and Pharma (ICAIH 2026), UHAS, Ghana (August 2026).

Presented: *Liora*, a private AI health companion designed to close Ghana’s HIV testing gap.

PROFESSIONAL EXPERIENCE

Data Analyst

Kumasi Center for Collaborative Research in Tropical Medicine

Oct 2025 – Present

Kumasi, Ghana

- Analyzed infectious disease and epidemiological datasets (SMART, PADMe, CLARITY, TreatPreg, POLARITY, MALDx) using R and spatial methods.
- Analyzed 1,200+ patient records on malaria-schizophrenia comorbidity; identified key epidemiological trends informing public health interventions.

Research Officer & Health Data Analyst (Contract)

Muslim Health Workers Association of Ghana

Jan 2024 – Dec 2024

Kumasi, Ghana

- Conducted free health screening of 500+ residents at Sawaba.
- Statistically analyzed medical diagnosis data using R, informing public health strategies.

LEADERSHIP, VOLUNTEERING & PROFESSIONAL ACTIVITIES

- Scientific Committee Member**, 1st GhES & 7th AfEA Conference (2025–Present). Abstract review, scientific program development, and session coordination.
- Standing Committee Member**, Ghana Epidemiological Society, Accra (2025–Present). Partnerships, health communication articles, and external relations.
- Regional Ambassador**, Centicini Research Team, Nigeria (2025–Present). Mentoring 50+ researchers in AI-powered systematic review and data analysis across 6 Ghanaian universities (team goal: 200,000 reviewers by 2030).
- Vice President**, Ghana Muslim Students Association – KNUST (2021–2022). Led a 40-member executive team; organized seminars, workshops, and sports; secured welfare packages in collaboration with NUGS-KNUST, KNUST SRC, and MUSLICOM KNUST.

HONORS AND AWARDS

- 500 USD Travel Grant, International Conference on AI in Healthcare and Pharma, UHAS, Ghana (2026)
- Research Grant Recipient, World Veterinary Association (2025)
- Dean’s Award for Outstanding Teaching Assistant, Obuasi Campus, KNUST (2023)
- Runner-up, Prof. Amekudzi Interdepartmental Quiz, College of Science, KNUST (2021)

TECHNICAL SKILLS

Python, R, MATLAB, GIS, STATA, L^AT_EX, SQL, Tableau, Power BI, Excel, REDCap, SPSS